

Testing and Data Analysis File

Methodology and Evaluation of the Data Analysis

This capstone analysis applies appropriate data preparation, exploratory techniques, and predictive modelling to a large population health dataset, using sound methodological choices including Winsorisation, non-parametric correlation, multinomial and binary logistic regression, and random forest modelling. However, model performance is fundamentally constrained by severe class imbalance, particularly for prediabetes, which led most models to favour the majority “no diabetes” class and struggle to reliably distinguish between classes, even when resampling and class-based weighting were applied. Reframing the task as binary risk identification did produce more stable and interpretable results, supporting the conclusion that the model is better suited to population-level screening rather than diagnostic classification.

Initial EDA: Summary

- There are profiles for 253,680 individuals in this dataset.
- The datafile has the following profile for the target variable Diabetes_012 with counts as follows:
 - no diabetes: 213,703 (84%)
 - prediabetes: 4,631 (1.8%)
 - diabetes: 35,346 (13.9%)
- There are an additional 21 variables, all of which are floats.
- The description of each feature is included in the README file (i.e. the feature name, the question asked and the possible response codes).
- The dataset has been cleaned.
- There are no null values.
- However, Body Mass Index (BMI) has both upper and lower outliers that need addressing.

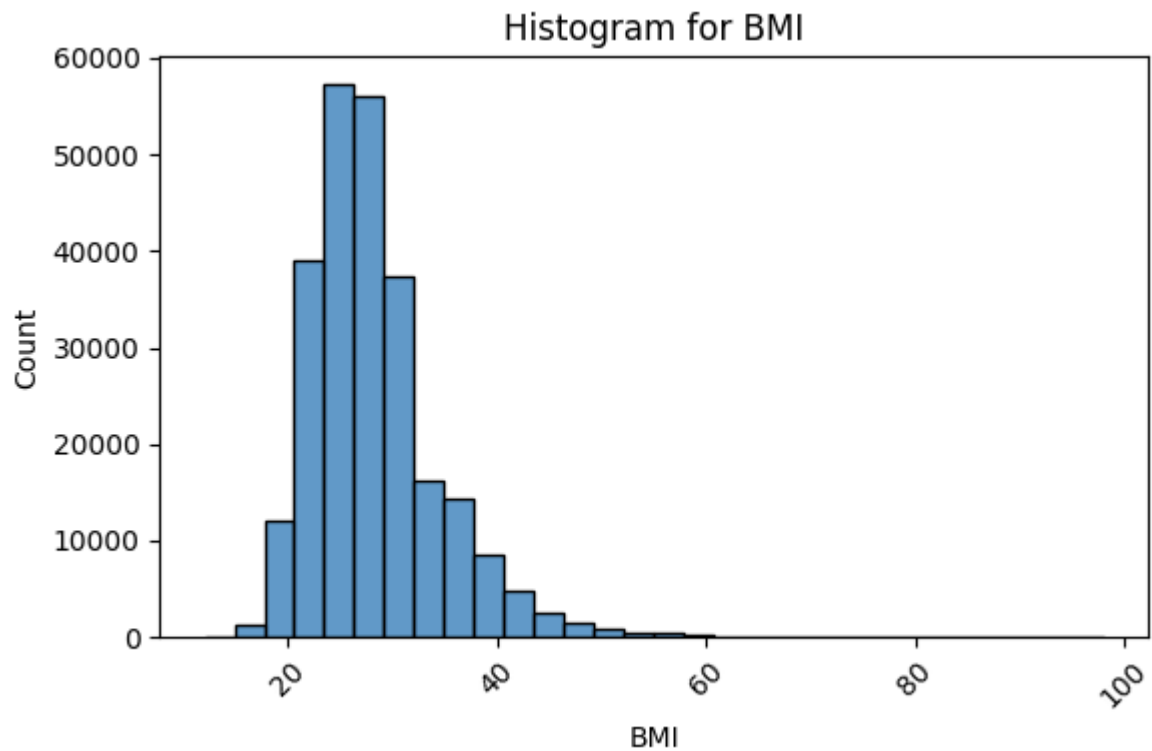
Data Analysis

The following table describes the original BMI values, while the histogram below offers a visual representation of their distribution.

Online research suggests that BMI is more typically between 18 and 50 (e.g. https://www.cdc.gov/healthyweight/assessing/bmi/adult_bmi/index.html (cdc.gov in Bing))

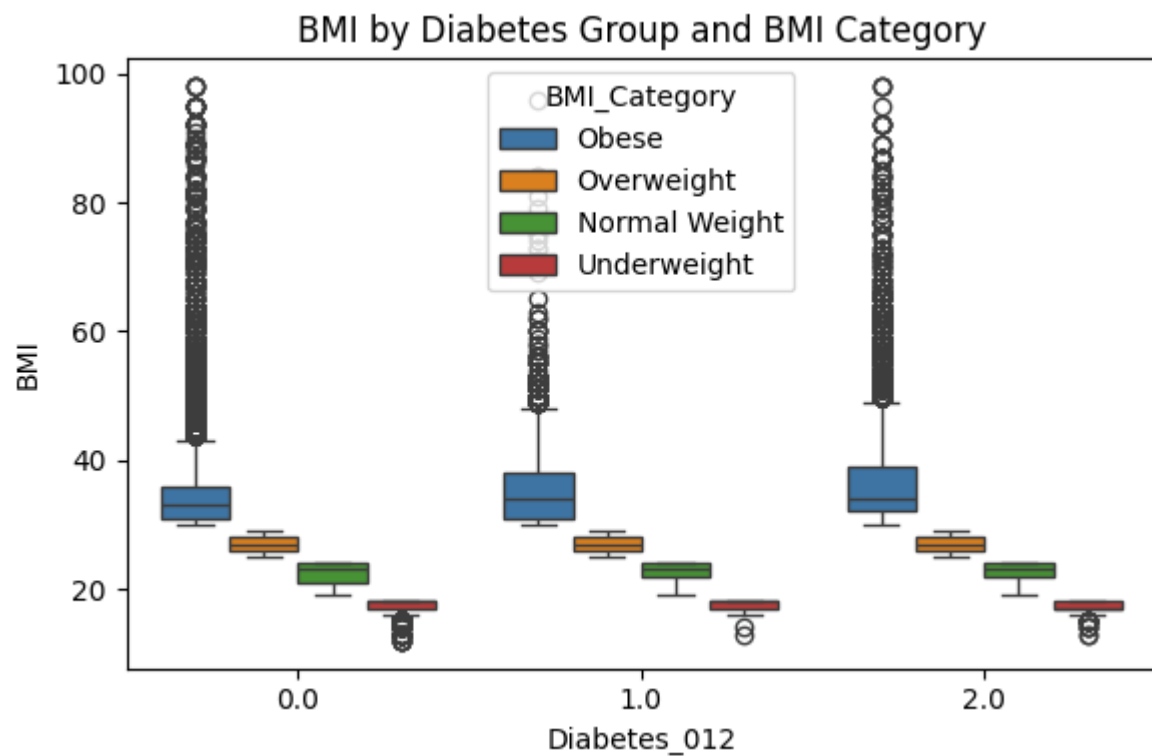
Therefore, in this dataset, there are some extreme outliers at both the lower and upper ends.

Variable	Count	mean	std	min	25%	50%	75%	max
BMI	253680	28	6.61	12	24	27	31	98



BMI was also grouped into 4 classifications to aid analysis:

- Underweight: <18
- Normal weight: >18 and ≤ 24.8
- Overweight: ≥ 25 and ≤ 29.9
- Obese: ≥ 30



Winsorisation

Description of Winsorisation

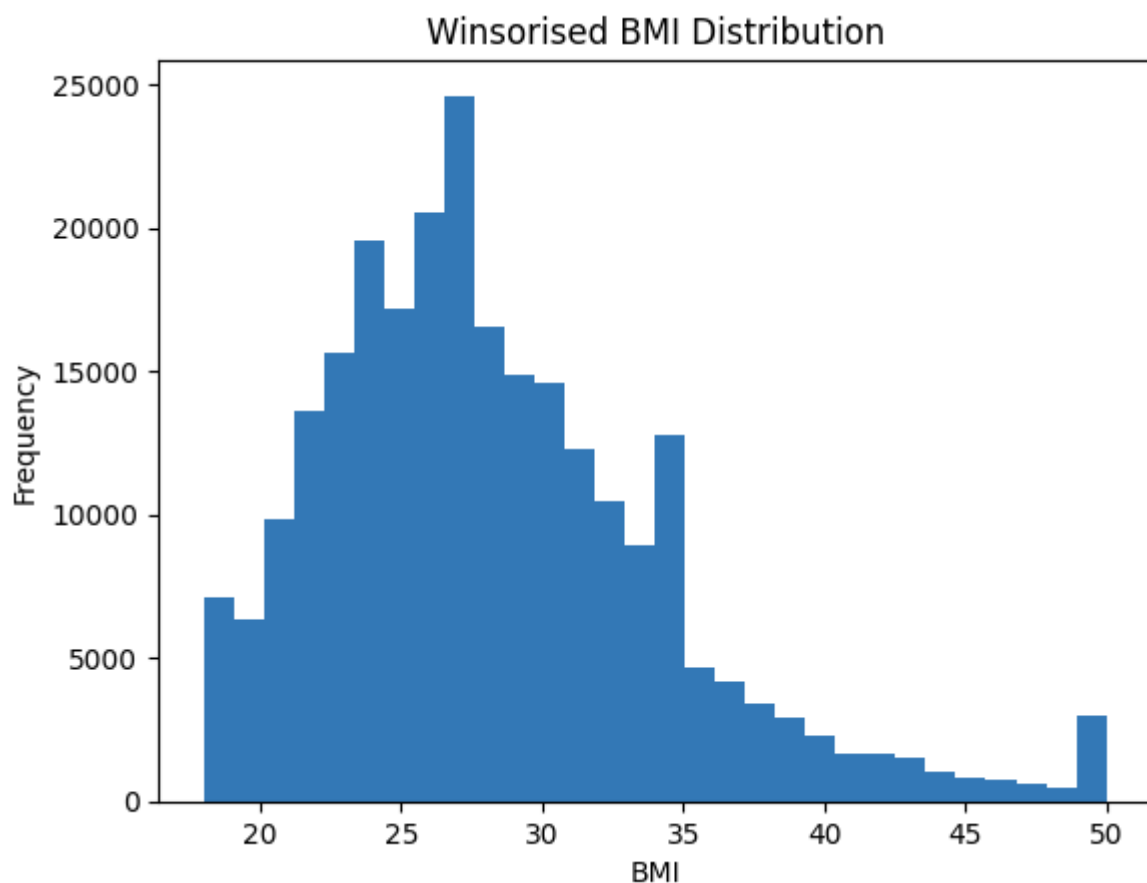
Winsorisation is a practical way to limit the influence of extreme values without discarding data entirely. Instead of removing outliers, they are capped at the chosen percentile thresholds. This preserves sample size and distributional shape while preventing extreme values from dominating visualisations or summary statistics.

Data Cleaning Undertaken

In this dataset, Winsorisation for BMI was set to 1% (BMI = 18) and 99% (BMI = 50). The revised BMI distribution is summarised in the following histogram.

Variable	Count	mean	std	min	25%	50%	75%	max
BMI_winsorised	253680	28	6.03	18	25	27	31	50

This variable, **BMI_winsorised**, will be used in all analyses going forward and BMI will be dropped.



Spearman Correlation

Description of Spearman Correlation

Spearman correlation is a rank-based measure of association that captures how well the relationship between two variables can be described by a monotonic trend (i.e. as one variable increases, the other always tends to decrease) rather than a straight

line. It is especially useful when the data is skewed, ordinal, or contains outliers, which is why it works so well in health and behavioural datasets like this one.

Analysis Undertaken

A Spearman correlation, rather than a Pearson correlation, was selected to support the analysis as the variable distributions were typically skewed (i.e. not normally distributed).

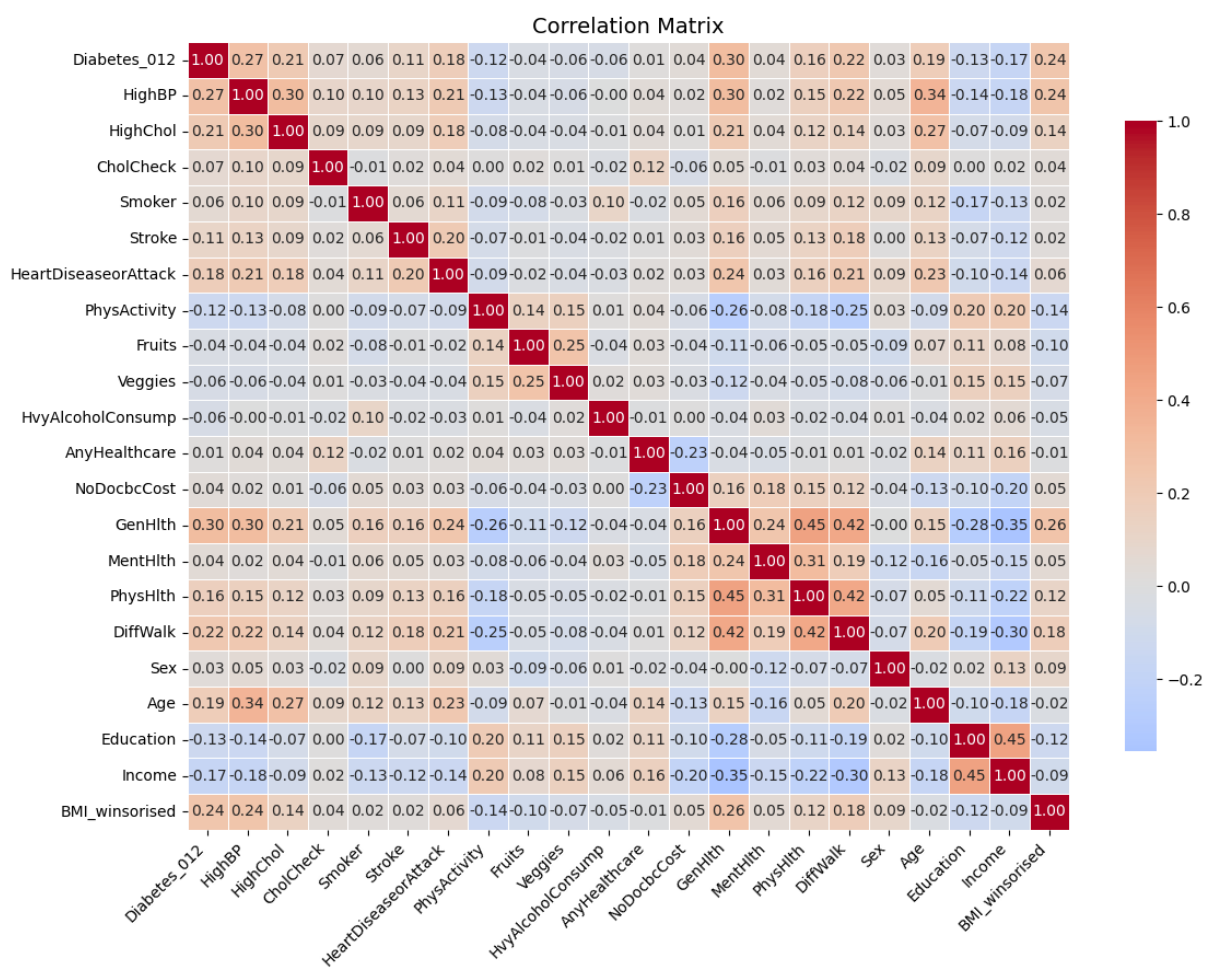
Two separate correlation analyses were conducted:

- An initial correlation of all features vs all features, to understand the relationships between features.
- A second correlation using Diabetes_012 as the target/outcome variable versus the remaining features.

Initial Correlation: All features vs All features using a Heatmap.

Findings

This initial correlation matrix shows how health and demographic variables are interrelated. General health is linked to physical health and mobility issues, while age is correlated with higher blood pressure and poorer overall health. A higher BMI is associated with increased cardiovascular risks, and lower income and education levels tend to be associated with worse health outcomes.



Second correlation: Diabetes_012 as the target variable

Findings

Once again, diabetes risk is driven most strongly by overall health, blood pressure, BMI, mobility limitations, and cholesterol, with age and heart disease also playing meaningful roles. Protective factors such as physical activity, higher education, and higher income show a lower likelihood of diabetes risk.

Feature Importance in Predicting Diabetes

1. Diabetes_012	1.000000
2. GenHlth	0.297138
3. HighBP	0.271668
4. BMI_winsorised	0.235914 (like original BMI)
5. DiffWalk	0.223567
6. HighChol	0.210668
7. Age	0.186357
8. HeartDiseaseorAttack	0.178564
9. PhysHlth	0.161718
10. Stroke	0.105887
11. CholCheck	0.068018
12. Smoker	0.063040
13. MentHlth	0.044921
14. NoDocbcCost	0.037379
15. Sex	0.030143
16. AnyHealthcare	0.014530
17. Fruits	-0.042268
18. HvyAlcoholConsump	-0.057244
19. Veggies	-0.059353
20. PhysActivity	-0.121988
21. Education	-0.126862
22. Income	-0.172611

Chi-square

Description of Chi-square

Chi-square is a statistical test that measures whether two categorical variables are associated. It compares the observed counts in a contingency table with the expected counts that you would get if there were no relationship between the variables.

Analysis Undertaken

Chi-square tests were undertaken for the 14 binary variables, with Diabetes_012 as the target variable.

Findings

The chi-square results show that all 14 variables have statistically significant associations with diabetes status, but the strength of those relationships varies

widely. High blood pressure, difficulty walking, high cholesterol, and heart disease/attack exhibit the strongest associations (highest Cramer's V), suggesting they are the most influential correlates of diabetes in this set.

Variables such as sex and (lack of) access to healthcare show very weak associations, indicating they have a minimal practical relationship despite statistical significance driven by the large sample size.

The results are shown in the table below.

Variable	Chi2	p	cramers_v
HighBP	18794.64	0.0	0.27219
DiffWalk	12776.94	0.0	0.22442
HighChol	11258.92	0.0	0.21067
HeartDieaseorAttack	8244.89	0.0	0.18028
PhysActivity	3789.30	0.0	0.12222
Stroke	2916.75	0.0	0.10723
CholCheck	1173.75	1.3291e-255	0.06802
Smoker	1010.51	3.7167e-220	0.06311
Veggies	893.84	8.0296e-195	0.05936
HvyAlcoholConsump	850.32	2.2619e185	0.05789
Fruits	454.347	2.1867e-99	0.04232
NoDocbcCost	396.082	9.8158e-87	0.03951
Sex	250.851	3.3767e-55	0.03145
AnyHealthcare	69.078	9.9979e-16	0.01650

Modelling

Model 1: Multinomial Logistic Regression

Description of Multinomial Logistic Regression

Multinomial logistic regression was used to model the probability of belonging to one of three outcome categories (i.e. no diabetes, prediabetes, or diabetes) based on the predictor variables. This method extends binary logistic regression by allowing the dependent variable to have more than two unordered categories, while still using the logistic function to estimate category-specific probabilities.

Analysis Undertaken

An initial multinomial regression was conducted using the original three-category outcome variable, retaining the codes:

- **No diabetes:** 213,703 individual responses
- **Prediabetes:** 35,346 individual responses
- **Diabetes:** 4,631 individual responses

This allowed the model to estimate how each predictor influenced the likelihood of membership in each category.

Definitions

Definitions of:

- **precision** = Of all the cases the model said were diabetes, how many were truly diabetes (i.e. few false positives)
- **recall** = Of all the people who have diabetes, how many did the model correctly identify? (i.e. few false negatives)
- **f1_score**: = How well does the model perform when we care about both catching true cases and avoiding false alarms?
- **support**: count of individuals in each class

In health analysis, **recall** is more typically the primary evaluation metric.

Findings

- Despite achieving moderate overall accuracy (69%), the multinomial logistic regression performs poorly on the two minority classes, especially prediabetes, indicating a limited ability to differentiate between the three diabetes classes.
- Individuals who are younger, leaner, healthier, and socioeconomically advantaged are much more likely to be classified as having no diabetes.

Classification report

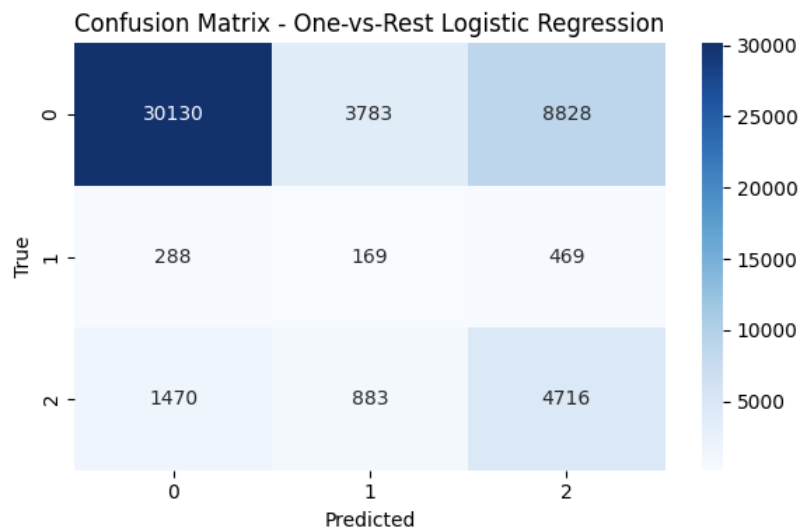
The model accuracy was 69%.

Variable: Class	precision	recall	f1-score	Support
No diabetes (0)	0.9449	0.7049	0.8075	42741
Prediabetes (1)	0.0350	0.1825	0.0587	926
Diabetes (2)	0.3365	0.6671	0.4474	7069
Accuracy			0.6901	50736
macro avg	0.4388	0.5182	0.4378	50736
weighted avg	0.8435	0.6901	0.7436	50736

Model 1: Multinomial Logistic Regression Confusion Matrix

No diabetes (class 0) shows that whilst most predictions are correctly identified (30,130), a further 12,611 are misclassified, especially diabetes (class 2). Furthermore, prediabetes (class 1) shows very poor performance with only 169 out of 926 cases correctly identified. Most are misclassified as diabetes (class 2). Finally, diabetes (class 2) shows moderate performance with 4,716 correct predictions out of 7,069, but with 1,470 misclassified as no diabetes (class 0).

In summary, the model over-predicts diabetes and misclassifies a substantial number of diabetes cases as healthy. This limits its usefulness in a medical setting.



Top positive & negative coefficients per class:

No diabetes (class 0): top positive features:

- HvyAlcoholConsump 0.145355
- Income 0.133858
- PhysHlth 0.055354
- Education 0.042370
- Fruits 0.024307
- PhysActivity 0.016320
- MentHlth 0.015768
- Veggies 0.011974

No diabetes (class 0): top negative features:

- GenHlth -0.586325
- BMI_winsorised -0.503940
- Age -0.467266
- HighBP -0.333542
- HighChol -0.293158
- CholCheck -0.228686
- Sex -0.128226
- HeartDiseaseorAttack -0.066704

Prediabetes (class 1): top positive features:

- Age 0.390335
- BMI_winsorised 0.310915
- GenHlth 0.264090
- HighChol 0.242938
- CholCheck 0.124227

- HighBP 0.122307
- NoDocbcCost 0.099967
- MentHlth 0.066339

Prediabetes (class 1): top negative features:

- Income -0.128164
- Education -0.060001
- DiffWalk -0.054067
- PhysHlth -0.049724
- HeartDiseaseorAttack -0.043618
- Stroke -0.040096
- AnyHealthcare -0.019065
- HvyAlcoholConsump -0.018618

Diabetes (class 2): top positive features:

- GenHlth 0.617056
- BMI_winsorised 0.505217
- Age 0.463535
- HighBP 0.357944
- HighChol 0.285405
- CholCheck 0.246751
- Sex 0.142219
- HeartDiseaseorAttack 0.074738

Diabetes (class 2): top negative features:

- HvyAlcoholConsump -0.166793
- Income -0.124718
- PhysHlth -0.057332
- Education -0.036977
- MentHlth -0.030262
- Fruits -0.025697
- PhysActivity -0.016740
- Veggies -0.011804

Model 2: Logistic Regression with a simple binary variable

1. Analysis Undertaken

A logistic regression was conducted with the target variable reduced to 2 outcome codes:

- Diabetes (code 0): 35,346 individual responses
- Prediabetes (code 1): 4,631 individual responses

The aim was to understand the factors that lead from prediabetes to diabetes.

Findings

This logistic regression comparing prediabetes with diabetes showed limited ability to distinguish between the two groups, achieving only moderate performance (ROC AUC = 0.62) despite the large sample size. Indeed, the performance for identifying prediabetes was particularly weak, reflecting both the severe class imbalance and the close similarity between the two metabolic states.

Overall, the available variables do not provide enough differentiation to reliably separate prediabetes from diabetes.

Findings

The logistic regression model demonstrated moderate overall performance, achieving an accuracy of 59% and a ROC AUC of 0.624, indicating limited ability to distinguish between prediabetes and diabetes.

Classification report

The model accuracy was 59%. ROC AUC was 0.6241.

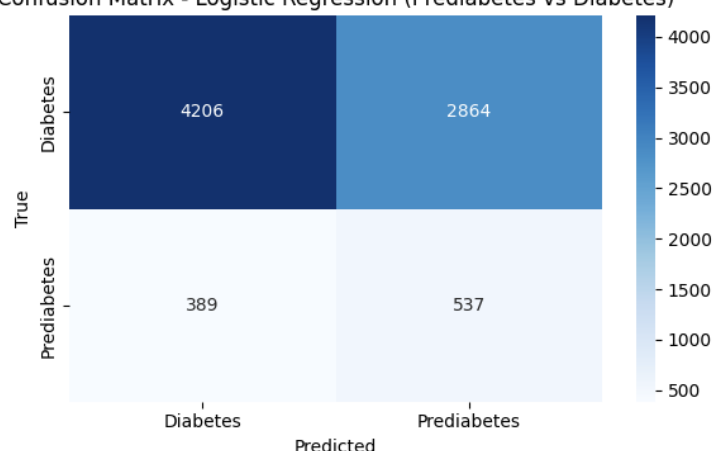
Variable: Class	precision	recall	f1-score	Support
Diabetes (0)	0.9153	0.5949	0.7211	7070
Prediabetes (1)	0.1579	0.5799	0.2482	926
Accuracy			0.5932	7996
macro avg	0.5366	0.5874	0.4847	7996
weighted avg	0.8276	0.5932	0.6664	7996

Model 2: Logistic Regression Confusion Matrix

The model correctly identifies 4,206 out of 7,070 cases, but 2,864 are misclassified as prediabetes. 537 out of 926 cases are correctly predicted as prediabetes, while 389 are misclassified as diabetes.

The model could be useful for initial screening but lacks the precision required for a formal diagnosis.

Confusion Matrix - Logistic Regression (Prediabetes vs Diabetes)



Prediabetes (class 1): top positive features:

NoDocbcCost	0.096627
HvyAlcoholConsump	0.095689
MentHlth	0.090416
PhysActivity	0.020658
PhysHlth	0.012075
Fruits	0.011002
Income	-0.001100
Smoker	-0.013267
Veggies	-0.022703
AnyHealthcare	-0.023970

Prediabetes (class 1): top negative features:

HighChol	-0.032735
Education	-0.037476
Age	-0.048128
DiffWalk	-0.056069
Sex	-0.077414
Stroke	-0.080416
HeartDiseaseorAttack	-0.114240
BMI_winsorised	-0.130547
HighBP	-0.166303
GenHlth	-0.283423

Random Forest

Description of Random Forest

A random forest is an ensemble machine-learning method that builds many decision trees and combines their predictions to produce a more stable, accurate result. Each tree sees a slightly different sample of the data and a random subset of features, which reduces overfitting and captures complex, non-linear relationships. Because of this structure, random forests tend to perform well on noisy, imbalanced, or high-dimensional datasets like health surveys.

Analysis undertaken with 3 and 2 classes.

Model 3: Random Forest with 3 Classes

An initial Random Forest was undertaken with 3 classes:

- 0 = No diabetes: 213,703 individual responses
- 1 = Prediabetes: 4,631 individual responses
- 2 = Diabetes: 35,346 individual responses

Findings

This model achieves an 84% accuracy, which is almost entirely driven by the 'no diabetes' class, i.e., the dominant class. It has extremely poor recall for prediabetes as well as for diabetes, which may be problematic in a real-world setting.

Feature importance is dominated by BMI, age, income, physical and general health, which are known diabetes risk factors.

Classification report

The model accuracy was 84%.

Variable: Class	Precision	recall	f1-score	Support
No diabetes (0)	0.8594	0.9696	0.9112	42741
Prediabetes (1)	0.0000	0.0000	0.0000	926
Diabetes (2)	0.4764	0.1597	0.2392	7069
Accuracy			0.8390	50736
macro avg	0.4452	0.3764	0.3835	50736
weighted avg	0.7903	0.8390	0.8009	50736

Model 3: Random Forest with 3 Classes Confusion Matrix

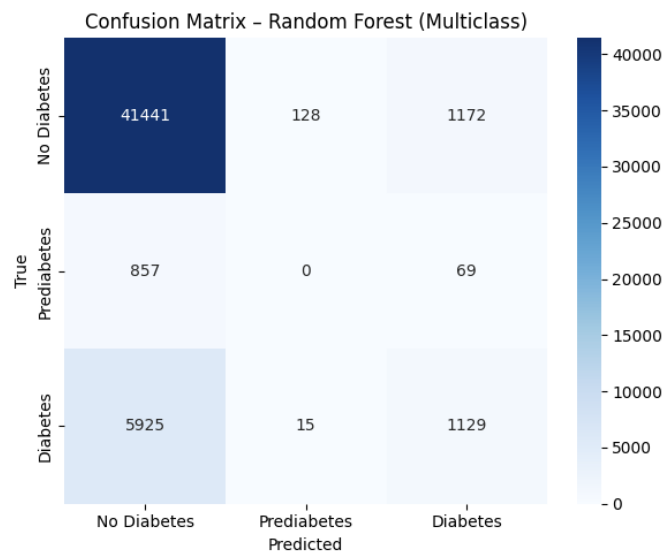
The model mostly correctly predicts no diabetes (41,441 predictions) but 1,300 cases are misclassified, mainly as diabetes (1,172 cases)

The model completely misses the prediabetes class, mostly misclassifying as no diabetes (857 cases) which suggests that it may not be able to make a meaningful decision for this category.

There is a diabetes underprediction, with the model heavily favouring predicting 'no diabetes,' even for true diabetes cases (5,925 cases), which could be extremely problematic.

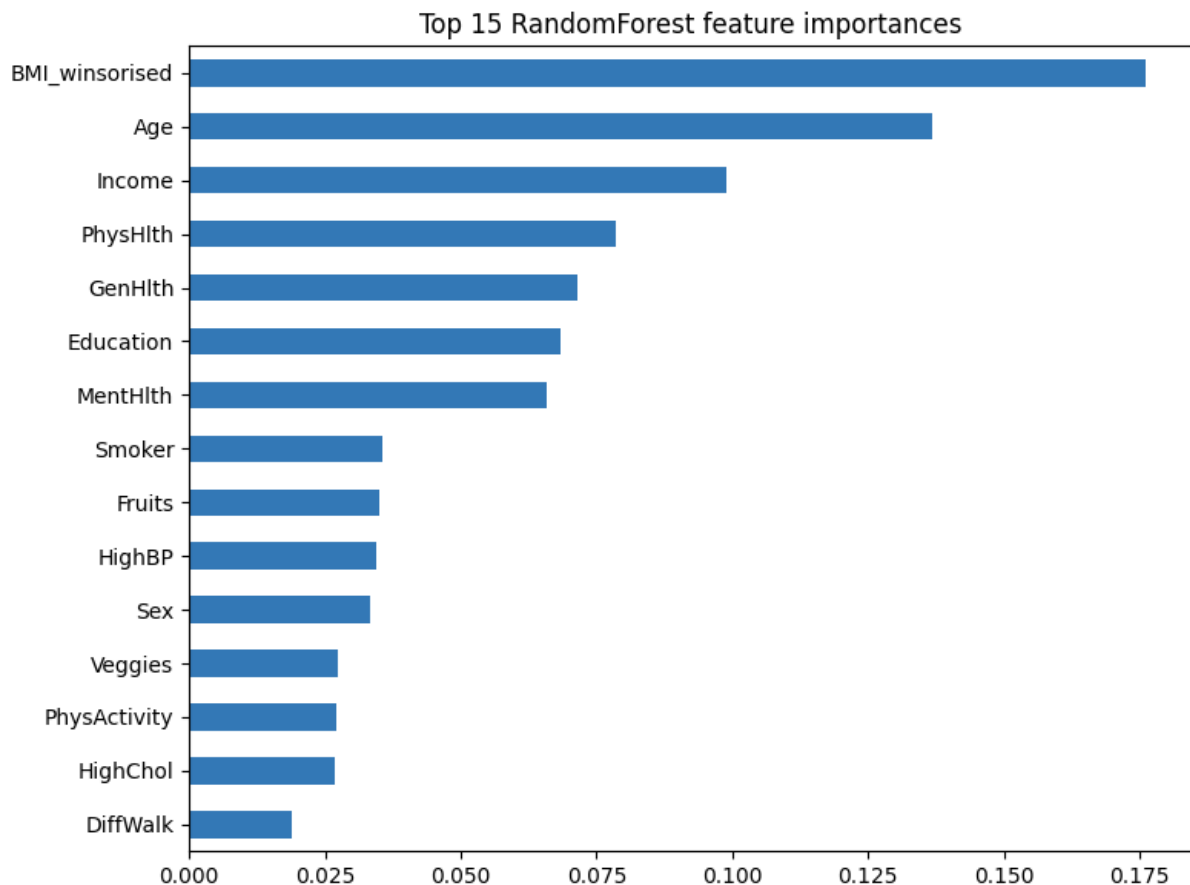
A key reason for this is likely to be the imbalance in the dataset with the dominance of 'no diabetes' predictions, which may be skewing the model's decision-making.

The model's failure to reliably distinguish the high-risk groups means that it will have limited usefulness in a real-world setting.



Top 15 feature importances:

- BMI_winsorised 0.176090
- Age 0.136864
- Income 0.098828
- PhysHlth 0.078618
- GenHlth 0.071413
- Education 0.068487
- MentHlth 0.065723
- Smoker 0.035564
- Fruits 0.035047
- HighBP 0.034379
- Sex 0.033271
- Veggies 0.027300
- PhysActivity 0.027211
- HighChol 0.026885
- DiffWalk 0.018963



Improving the Model

To improve the model, two further actions were undertaken:

- SMOTE oversampling to increase representation of minority classes.
- Class-based weighting

SMOTE

Definition of SMOTE

SMOTE is a resampling technique used to fix class imbalance by creating synthetic minority-class examples rather than simply duplicating existing ones. It works by selecting a minority sample, finding its nearest neighbours, and generating new points along the line between them, which helps the model learn a fuller decision boundary instead of being overwhelmed by the majority class.

Results

SMOTE did not improve the model. Its findings are not included.

Class-Based Weights

Definition of Class-Based Weights

Class-based weights are a mechanism used in machine learning models to compensate for imbalanced datasets like this one by giving more influence on the underrepresented classes during training.

When one class dominates (e.g. the 'no diabetes' class represents 84% of the dataset), a model can achieve high accuracy simply by predicting the majority class. Class-based weights counteract this by:

- Increasing the penalty for misclassifying minority-class samples
- Reducing the penalty for misclassifying majority-class samples
- Shifting the model's decision boundaries so it pays more attention to the minority class.

Model 4: Random Forest with 2 Classes and class-based weighting

A second Random Forest model was undertaken with 2 classes:

- 0 = No diabetes: 213,703 individual responses
- 1 = Prediabetes/Diabetes: 39,977 individual responses

Findings

The model achieved 74% accuracy. Once again, the model was strong at identifying individuals without diabetes while still capturing the most at-risk cases despite the class imbalance. Feature importance results highlight that overall health status, blood pressure and BMI are the dominant predictors for diabetes.

Classification report

The model accuracy was 74%.

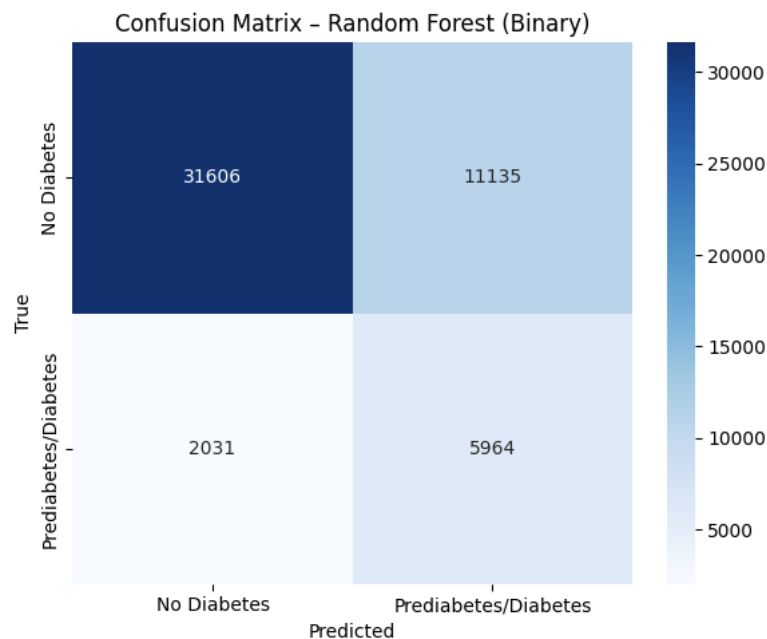
Variable: Class	precision	Recall	f1-score	Support
No diabetes (0)	0.9396	0.7395	0.8276	42741
Prediabetes/Diabetes (1)	0.3488	0.7460	0.4753	7995
Accuracy			0.7405	50736
macro avg	0.6442	0.7427	0.6515	50736
weighted avg	0.8465	0.7405	0.7721	50736

Model 4: Random Forest (2 Classes and Class-Based Weighting) Confusion Matrix

Recall is the most important metric in this diagnostic setting, and it is strong and similar for both classes (i.e. no diabetes and prediabetes/diabetes), with the model identifying circa 75% of cases.

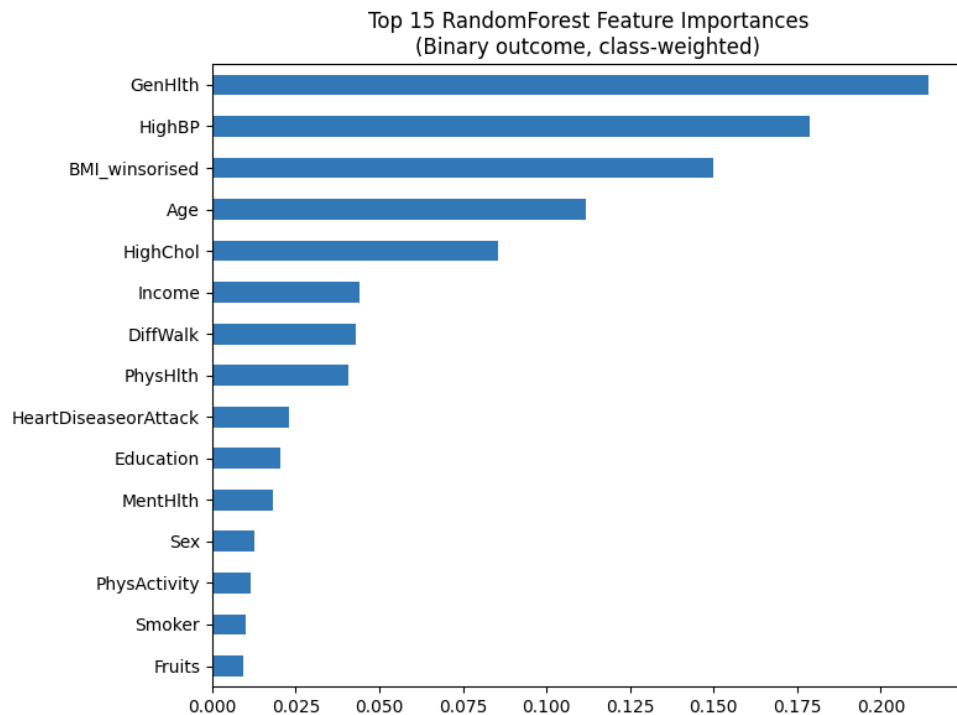
However, the precision is not so good as 11,135 healthy individuals are incorrectly flagged as having a diabetes risk. Equally, 2,031 individuals who have pre/diabetes are incorrectly flagged as no diabetes.

This means in a real-world setting, the model may be useful for initial screening, but it is not reliable for confirming a formal prediabetes or diabetes diagnosis.



Top 15 feature importances:

- GenHlth 0.214270
- HighBP 0.178877
- BMI_winsorised 0.149961
- Age 0.111840
- HighChol 0.085582
- Income 0.044127
- DiffWalk 0.042807
- PhysHlth 0.040593
- HeartDiseaseorAttack 0.022892
- Education 0.020479
- MentHlth 0.018176
- Sex 0.012761
- PhysActivity 0.011353
- Smoker 0.010054
- Fruits 0.009378



These top 15 features were then grouped into 1 of 5 groups as follows:

Clinical / Physiological:

- GenHlth
- HighBP
- BMI_winsorised
- HighChol
- HeartDiseaseorAttack

Demographic:

- Age
- Sex

Functional Health:

- DiffWalk
- PhysHlth
- MentHlth

Socioeconomic:

- Income
- Education

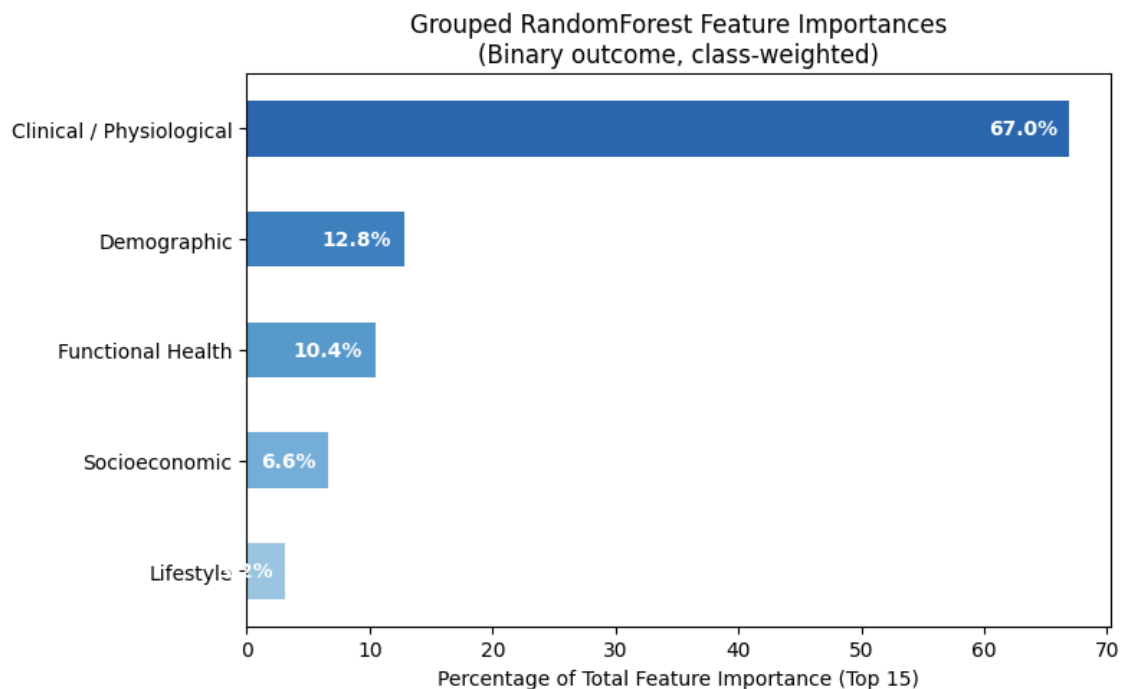
Lifestyle:

- Smoker
- Fruits

- PhysActivity.

Groups:

- Clinical/Physiological: 67%
- Functional health: 12.8%
- Demographic: 10.4%
- Socioeconomic: 6.6%
- Lifestyle: 3.2%



Hyperparameter Tuning for Random Forest

Hyperparameter tuning for Random Forest controls how the forest learns to provide better predictions, better generalisations, and more stable models. Random Forest already reduces over-fitting by averaging many trees. Hyperparameter tuning fin-tunes the bias-variance trade-off.

Model 5: Random Forest Model with Hyperparameter Tuning

A final Random Forest model with hyperparameter tuning was generated with 2 classes:

- 0 = no diabetes: 213,703 individual responses
- 1 = prediabetes/diabetes: 39,977 individual responses

Please Note

This element of the model led to the computer crashing and hanging on numerous occasions. However, it does perform best, and this is the model used in the presentation and dashboard.

Findings

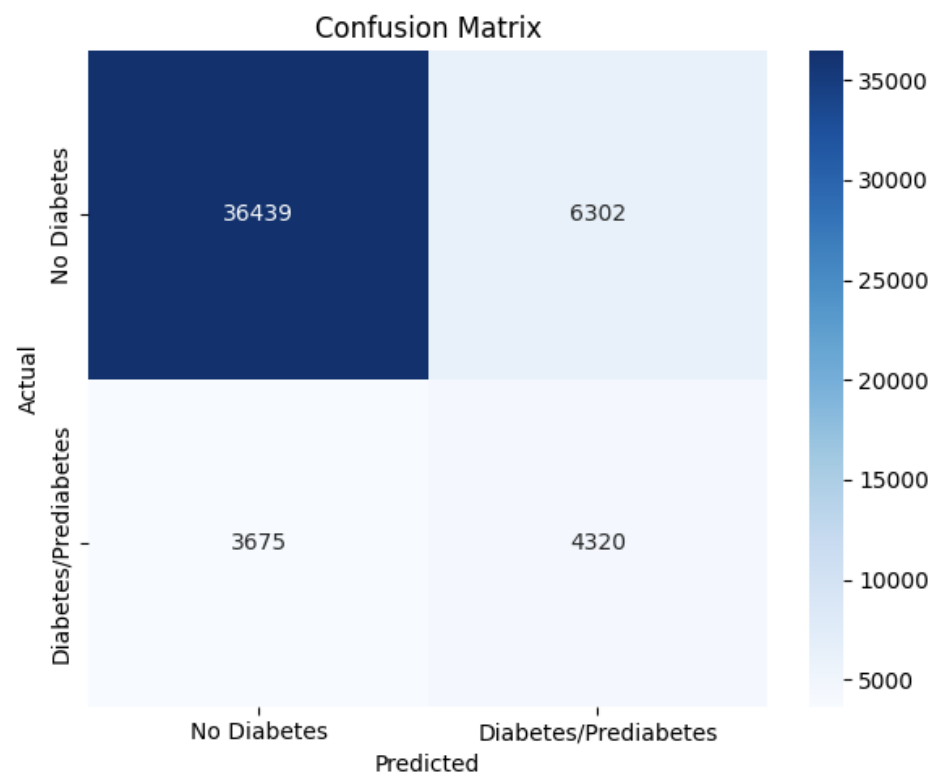
The model achieves a recall of 55% for the prediabetes/diabetes class, correctly identifying 4,320 individuals with elevated risk. However, the precision for this class is 41%, indicating that many individuals flagged as “at risk” do not actually have diabetes (6,302). At the same time, the model misses 3,675 true cases, predicting “no diabetes” for individuals who do in fact have the condition.

The model has value as a broad early-stage screening tool, but its performance is not strong enough for diagnostic confirmation.

Classification report

The model accuracy was 80%.

Variable: Class	precision	Recall	f1-score	Support
No diabetes (0)	0.9084	0.8525	0.8796	42741
Prediabetes/Diabetes (1)	0.4067	0.5483	0.4641	7995
Accuracy			0.8034	50736
macro avg	0.6575	0.6964	0.6718	50736
weighted avg	0.8293	0.8034	0.8141	50736



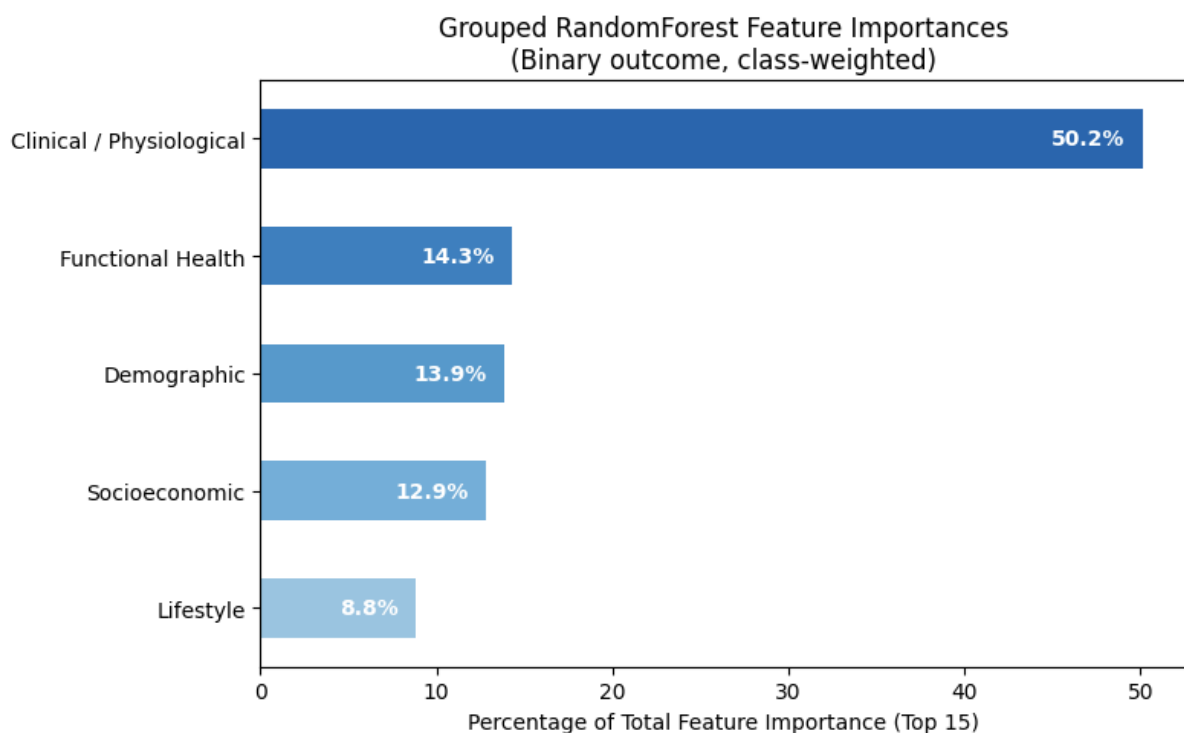
All feature importance:

- BMI_winsorised: 0.156779
- GenHlth: 0.145500
- Age: 0.117952
- HighBP: 0.107613
- Income: 0.068414

- PhysHlth: 0.064184
- HighChol: 0.057104
- MentHlth: 0.048144
- Education: 0.043550
- DiffWalk: 0.030598
- Smoker: 0.021375
- Fruits: 0.021355
- Sex: 0.020849
- HeartDiseaseorAttack: 0.019409
- PhysActivity: 0.019118
- Veggies: 0.017157
- NoDocbcCost: 0.010246
- HvyAlcoholConsump: 0.009233
- Stroke: 0.007801
- CholCheck: 0.007309
- AnyHealthcare: 0.006308

The model built the following groups:

- Clinical/Physiological: 50.2%
- Demographic: 13.9%
- Functional health: 14.3%
- Socioeconomic: 12.85%
- Lifestyle: 8.8%



SHAP

SHAP (SHapley Additive exPlanations)

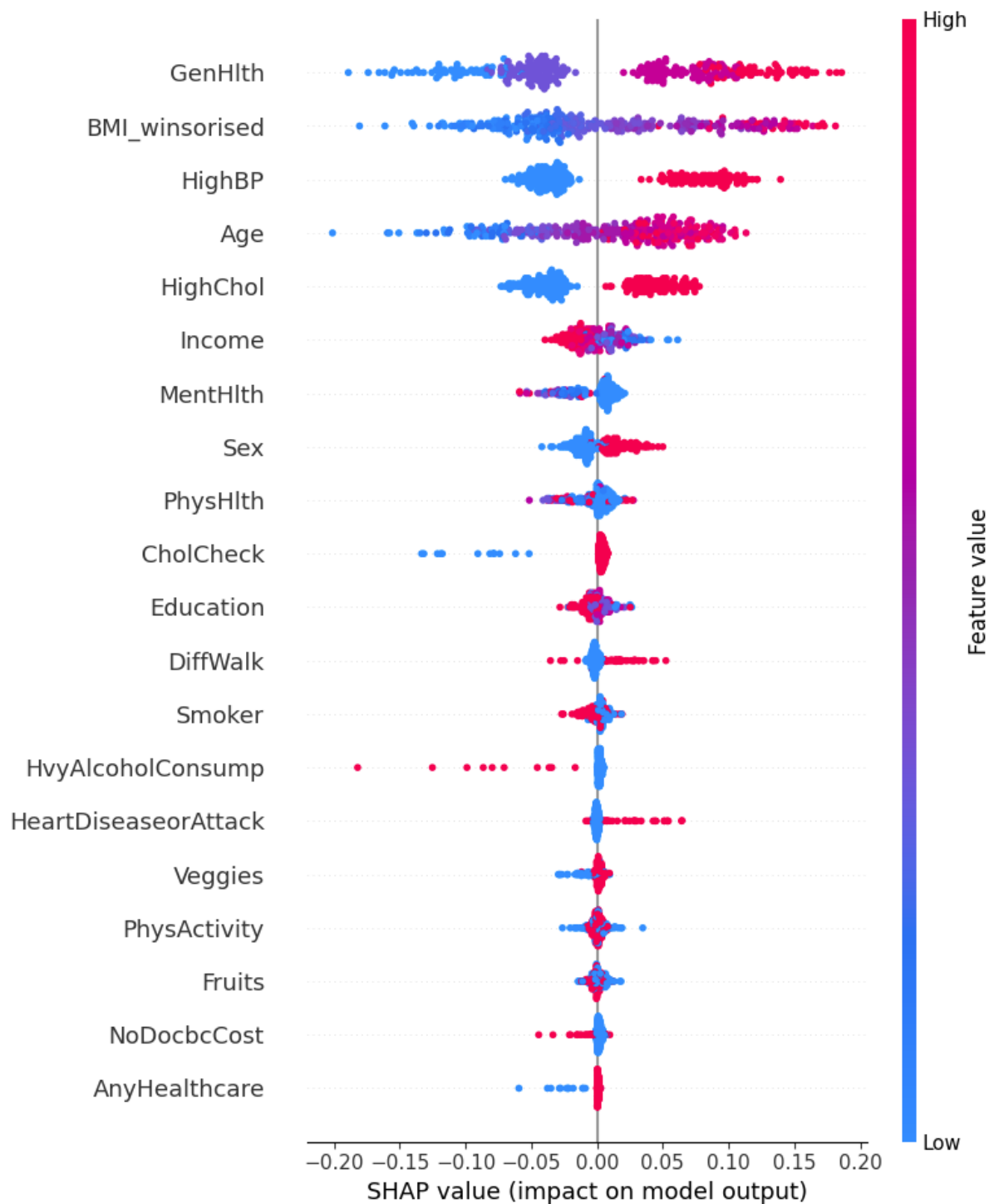
A method that shows how much each feature pushes a prediction up or down, using ideas from cooperative game theory.

Why it matters

- It tells you which features matter most.
- It shows whether a high or low value increases risk.
- It gives transparent, model-agnostic explanations for complex models like Random Forests or XGBoost.

Findings

The SHAP summary shows that **clinical health measures** (general health, BMI, blood pressure, cholesterol) were the strongest drivers of diabetes-risk predictions, **demographics** (age and income) were the next most influential, **functional limitations** added meaningful but smaller impact, and **lifestyle behaviours** contributed the least compared with the other factor groups.



Further Research Opportunities

Feature-level investigations:

- Partial dependence analysis to understand how individual predictors influence predicted diabetes risk.
- SHAP value decomposition to provide model-agnostic explanations and highlight non-linear or interaction effects.

- Interaction term exploration within logistic models to test whether combinations of variables meaningfully shift risk.

Alternative modelling strategies:

- Gradient boosting approaches (e.g., XGBoost, LightGBM, CatBoost) which often handle imbalance and non-linear structure more effectively than random forests.
- Calibration analysis to assess whether predicted probabilities reflect true risk, especially important in screening contexts.

Class imbalance–focused techniques:

- Synthetic data generation using SMOTE variants (SMOTE-NC, Borderline-SMOTE) tailored to mixed data types.
- One-class classification to model prediabetes as an anomaly detection problem rather than a multi-class task.
- Threshold optimisation to tune decision boundaries for sensitivity vs specificity trade-offs.